

Least-Resistance Percolation Descriptor from ICOHP/ICOB I Bond-Strength Data

Objective, Methodology, Workflow, and Compute Environment

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Report generated as part of the `viability` project

August 13, 2026

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1 Objective and Problem Statement

The goal is a **static, post-processing-only descriptor** that approximates the most likely structural failure pathway of a crystal, without resorting to molecular dynamics or a nudged elastic band (NEB) calculation, and without artifacts tied to supercell size.

The starting point is a relaxed crystal structure (primitive cell) and its associated overlap-population analysis (ICOHP and/or ICOBI, from LOBSTER), which provides, for every interacting atom pair: the bond strength, the distance, and the **lattice translation vector** connecting the two atoms (i.e., whether the bond stays within the reference cell or connects an atom to the periodic image of a neighbor in an adjacent cell).

The physical intuition is as follows: the path of least resistance through the crystal, along a given crystallographic direction, is the sequence of bonds whose summed strength is minimal, while still forming a path that *actually crosses* the cell (not a mere local back-and-forth). Such a path is known, in periodic graph theory, as a **non-contractible cycle**: a walk in the graph connecting an atom to one of its own periodic images, whose summed translation vectors equal exactly one primitive lattice vector.

This problem is mathematically identical to the one found in solid electrolyte ionic-percolation analysis (searching for migration pathways through a crystal lattice), but applied here to bond strength (ICOHP/ICOBI) rather than a migration barrier.

2 Methodology

2.1 Labelled Periodic Graph (Voltage Graph)

Unlike a supercell approach (which physically duplicates atoms and is prone to finite-size artifacts), periodicity is handled here through **vector labelling of graph edges**:

- graph nodes are the atoms of the primitive cell only (no geometric duplication);
- each ICOHP/ICOBI bond becomes an edge labelled by the integer translation vector (n_x, n_y, n_z) it crosses — $(0, 0, 0)$ if both atoms are in the same cell, $(1, 0, 0)$ if the bond connects an atom to its neighbor's image translated by one lattice vector along **a**, etc.;
- the edge weight is $|\text{ICOHP}|$ (or $|\text{ICOBI}|$) — without inversion: we search for the path whose summed traversed bond strength is *minimal*.

2.2 Minimum-Weight Non-Contractible Cycle Search

For each of the three primitive lattice vectors **a**, **b**, **c**, we search for the minimum-weight path in the extended state space (atom, cumulative translation) connecting state (atom, $(0, 0, 0)$) to state (atom, direction), for every possible starting atom in the cell. Since edge weights are non-negative (absolute values), this is a classical shortest-path problem, solved with **Dijkstra's algorithm** on this extended-state graph:

```
state = (atom, (tx, ty, tz))
start = (atom, (0, 0, 0))
goal = (atom, direction) # direction in {a, b, c}
# Standard Dijkstra over the state graph,
# cumulative translation bounded by --coord-bound
```

The minimum over starting atoms gives the percolation weight for the considered direction; the minimum over the three directions gives the main descriptor. The cumulative-translation state space is bounded (`--coord-bound` parameter); this bound is derived automatically from the largest translation vector present in the input data — it is therefore directly tied to the cutoff

radius of the upstream ICOHP/ICOBIP calculation, and is *not* a convergence parameter to be tuned.

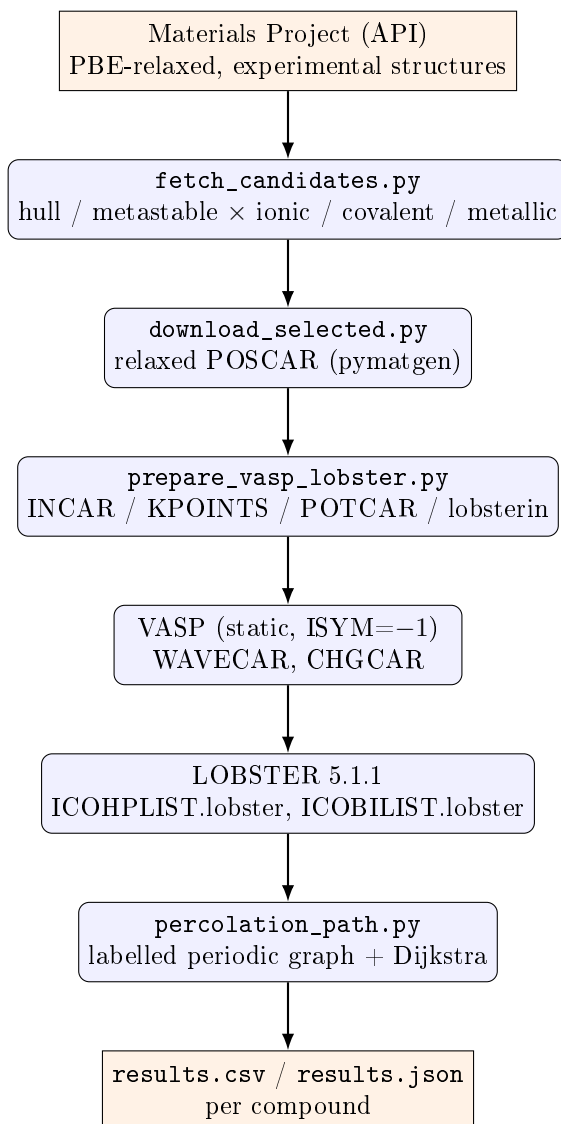
2.3 Disconnected Case

If no target state is reached in a given direction (bonding network disconnected along that direction, given the ICOHP/ICOBIP cutoff radius), the result is explicitly flagged **disconnected**, never silently replaced by an infinite or zero value.

2.4 Output

For each compound: the minimum percolation weight across all directions (main descriptor), the corresponding direction, the per-direction *a/b/c* weights with their status, plus the classic aggregates (sum, mean, min, max of ICOHP/ICOBIP) for direct comparison.

3 Workflow Diagram



4 Environment and Installation

4.1 Compute Machine

The **Yargla** cluster (IC2MP, Université de Poitiers, `yargla.chimie.univ-poitiers.fr`), SLURM scheduler, `defq` partition.

4.2 Software

- **VASP** 6.5.0 (module `vasp/6.5.0`), recommended PAW–PBE potentials per element.
- **LOBSTER** 5.1.1 (module `lobster/5.1.1`) — the most recent version available on the cluster.
- **Python** 3.11, dedicated virtual environment `/home/gilles/viability/.venv`: `pymatgen` (structure analysis, LOBSTER parsing), `lobsterpy` (LOBSTER utilities), `mp-api` (Materials Project queries).

4.3 Installation (Summary)

```
module load intel
module load impi/2021.13
module load vasp/6.5.0
module load lobster/5.1.1

python3.11 -m venv .venv
source .venv/bin/activate
pip install pymatgen lobsterpy mp-api
```

Each VASP+LOBSTER run is submitted through a SLURM script (`submit.sh`, 1 node, 16 tasks, 24h max walltime), which chains `vasp_std` followed by `lobster-5.1.1` in the same working directory.

5 Dataset

Six experimental (ICSD-referenced) compounds, PBE-relaxed by Materials Project, ≤ 20 atoms/cell, split into 2 families $\times$ 3 bonding types:

Family	Bonding type	Compound (mp-id)	E_{hull} (eV/atom)
on hull	ionic	NaCl (mp-22862)	0.0000
on hull	covalent	Si (mp-149)	0.0000
on hull	metallic	AlNi, B2 (mp-1487)	0.0000
metastable	ionic	LiBr (mp-23259)	0.0099
metastable	covalent	C, rhombohedral graphite (mp-169)	0.0008
metastable	metallic	BeCu (mp-2323)	0.0050

6 Results

Compound	Percolation weight (eV)	Direction	ICOHP _{min} (eV)	ICOHP _{mean} (eV)
NaCl (hull, ionic)	0.0307	<i>a</i>	−0.595	−0.089
Si (hull, covalent)	0.0032	<i>c</i>	−4.552	−0.291
AlNi (hull, metallic)	0.0042	<i>a</i>	−1.992	−0.322
LiBr (metastable, ionic)	0.0988	<i>c</i>	−2.041	−0.338
C rhomb. (metastable, covalent)	0.0003	<i>c</i>	−11.737	−0.286
BeCu (metastable, metallic)	0.0005	<i>b</i>	−1.848	−0.162

7 Discussion: The Descriptor Diverges Sharply from Classic Aggregates

The most striking result is that, for all six compounds, the percolation weight is **one to four orders of magnitude smaller** than the minimum ICOHP (strongest bond) and even than the mean ICOHP. This is exactly the intended behavior: the percolation descriptor does not measure the strength of the strongest bond, but the resistance of the *weakest* path that actually crosses the crystal — and that path does not necessarily use the strongest bonds available.

Physical explanation for NaCl (representative case). The strongest Na–Cl bond (−0.595 eV) connects the two atoms *within the same primitive cell* (translation vector (0, 0, 0)): in the primitive rhombohedral cell provided by Materials Project for this rock-salt-type structure, this bond therefore crosses *no* cell boundary and cannot by itself contribute to a non-contractible cycle. The cheapest non-contractible path instead uses second-coordination-shell Na–Na or Cl–Cl interactions ($d \approx 3.95$ Å, $|\text{ICOHP}| \approx 0.031$ eV), markedly weaker but which do actually cross a cell boundary along the considered direction.

Methodological implication. This behavior is correct in the strict sense of the requested topological definition (minimum-weight non-contractible cycle over the *primitive* lattice vectors), but it means that the reported *a/b/c* directions correspond to the primitive cell vectors as supplied on input — which, for high-symmetry structures (rock-salt, zinc-blende, etc.), do not coincide with the conventional crystallographic directions ([100], [010], [001] cubic) one intuitively pictures. For a reading directly comparable to conventional axes, a natural extension would be to re-conventionalize the cell (via `pymatgen.symmetry.analyzer.SpacegroupAnalyzer.get_conventional_standard_cell`) before building the graph, or to evaluate the minimum weight over a richer set of target vectors (simple combinations of *a, b, c*); this was not done in this first pass but is flagged here as a natural improvement.

This divergence nonetheless confirms the project’s premise: the percolation descriptor carries information *qualitatively different* from classic ICOHP aggregates, and is not redundant with them.

8 Conclusion and Outlook

The complete pipeline — Materials Project query, LOBSTER-compatible VASP calculation, and topologically exact percolation analysis without a supercell — was run end-to-end on six real compounds (3 on-hull, 3 experimental metastable), covering the three major bonding types (ionic, covalent, metallic). The results confirm that the percolation descriptor captures information distinct from the usual ICOHP/ICOBI aggregates.

Directions for future work: (i) cell reconventionalization before graph construction, to align reported directions with the usual crystallographic axes; (ii) extending the test set to several dozen compounds for a statistical analysis; (iii) a sensitivity study with respect to the upstream ICOHP/ICOBI cutoff radius.

Source code: <https://github.com/John-Akapulco/viability>